VeriLoom: Guidelines as an Executable Specification

Round 3 submission: The Finished Product

Finalist submission for the problem statement by Inkloom AI · Nexus Code

Item	Detail
Team name	nandhaashwin111222
Team leader	Nandha K
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Live project	https://veriloom-3d0g.onrender.com/
Source code	github.com/nandhak6197-a11y/VeriLoom (main branch)
Submission type	Single team-leader submission
Development status	Finished and deployed; 106 of 106 automated tests pass (100%)

Summary

The product. VeriLoom turns a brand guideline into something a machine can execute. Instead of wrapping one large prompt around a language model, it runs every draft through a nine-layer neuro-symbolic pipeline: deterministic compilers and exact formulas decide what they can, and a small locally hosted BERT-NLI model is used only where judgment is needed. The workbench has six core modules and a Copilot, with no placeholder screens and no stubbed APIs.

What was committed. At Round 2 we named three improvements, and these are the three delivered here, none dropped or swapped:

- **Time-bounded Fact Vault.** Each claim is checked against facts that are valid on the date of review, and against the strength of its wording, with surgical diffs for stale claims.
- **Multi-role governance and cryptographic ledger.** Four roles with permission gating, a reviewer override rate, and a SHA-256 hash chain that makes tampering visible.
- **Multimodal vision maths.** Colour distance (CIEDE2000) and WCAG 2.1 contrast are calculated, not estimated by a vision model.

Evidence. 106 of 106 automated tests pass across 12 modules. On the 18-piece Northwind golden benchmark (54 runs) violation detection reached F1 1.000, verdicts were identical across all three runs, and no high-risk hallucination was recorded. An 8-phase live end-to-end test produced 16 timestamped visual receipts. Section 9 gives the full figures.

Run it yourself. The README lets any reviewer clone, install and run the product locally in under 60 seconds, with no cloud dependency.

1. Problem recap and what Round 1 committed

Round 1 framed brand failure as three different problems, because each needs a different remedy:

- **Content errors:** a specific piece is wrong, off-brand or inaccessible.
- **Process failures:** a review exists but misses the error, or the content changes after sign-off.

- **Specification failures:** the guideline itself is unclear, outdated or contradictory, so reviewers cannot agree on what correct means.

The Round 1 thesis was that a brand guideline is a specification that is rarely tested, versioned or checked for ambiguity, and that a generic language-model reviewer gives different verdicts on the same input. In the author's own baseline test, 4 of 18 pieces (22.2%) changed verdict between three identical runs, and an expired fact was accepted as support. Those two findings shaped the design that follows.

1.1 Incidents that motivated the design

Incident	Failure type	Lesson for the design
Coca-Cola “Classic” campaign (2025): an ad appears to credit a book that does not exist	Factual error that passed review	Every claim must trace to an approved fact.
Equinox campaign image (2026): months of criticism before removal	Tone and inclusion; review failure	Reviewers need evidence and a risk-ranked queue.
Meta AI features and advertiser liability (2026)	Post-approval drift	Keep a fingerprint of what was approved (future scope).
Callaway and Good Good (2026): a video approved that should not have been	Approval failure; no risk routing	Route risky items to an Authorised Reviewer with a record.

Sources and verification status for these incidents are listed in the Round 1 submission (Appendices A and B).

2. From plan to product

Round 1 promised to report misses as well as hits. This section sets the Round 1 plan beside what the finished product does.

2.1 Where each Round 1 feature landed

Round 1 commitment	What the finished product has
Hedge-sensitive, time-aware claim verifier	Time-Bounded Fact Vault with [valid_from, valid_until] windows, strict, hedged and absolute modality classes, and modality-inflation flags (layers 4 and 5).
Intent-locked surgical rewriting	Surgical character diffing and one-click Accept Suggested Fix, scored by IPM, measured at 0.96 (layer 8).
Measured visual and accessibility checks	CIEDE2000 colour distance, WCAG 2.1 AA contrast, OCR legibility and logo aspect ratio (layer 6).
Four-role review workflow and tamper-evident audit log	Creator, Brand Manager, Authorised Reviewer and Brand Admin views, with a SHA-256 hash chain of every submission, waiver, approval and fact change (layer 9).
Reviewer Override Rate telemetry	Φ ROR compares reviewer overrides with automated findings to point at ambiguous clauses.
Verdict tiers and a stability gate	Four risk tiers (PASS, ADVISORY, VIOLATION, BLOCKED) with 100% identical verdicts across three runs (layer 7).
Privacy and injection defence	OWASP LLM01 injection defence and local PII masking with no external data transfer (layer 2).

2.2 The Round 1 hypotheses, H1 to H7

Round 1 set these targets before any measurement. The results column is filled only where this submission has a measured figure.

Hypothesis	Round 1 target	Result in the finished product
H1. Deterministic checks are precise on what they cover	Precision of at least 95%	Met: 100% precision and recall on the golden set (F1 1.000).
H2. Violation-tier findings are rarely overridden	Override rate below 20%	[add result]
H3. Grounded verdict tiers are more stable than free-form chatbot review	Better than the baseline (4 of 18 changed)	Met: 18 of 18 identical across three runs (100.0%).
H4. The verifier never accepts an expired fact	Zero cases	Expired fact F-DUR-01 is flagged and replaced by F-DUR-02. [add count of expired-fact test cases]
H5. Linting finds real guideline defects	At least 90% of seeded defects found	[add result]
H6. Intent Lock preserves must-keep points	100%	Met: IPM 0.96, with all protected words retained.
H7. Creators reach a clean submission faster	No target; report what is seen	[add result]

3. Compliance with the 72-hour competition mandate

The finalist round set six requirements. The table below states how each one was met.

Requirement	How it was met
1. Deliver exactly the three improvements named at Round 2	All three are complete: (1) Time-Bounded Fact Vault and Modality Locks, (2) Multi-Role Governance and Cryptographic Ledger, (3) Multimodal Vision Math (CIEDE2000 and WCAG 2.1). None was omitted or replaced.
2. Finish open features and close loose ends	The Copilot streams word by word with an active cursor. Chips use translucent frosted glass (rgba(255,255,255,0.08)), progress toasts last 10 seconds in metallic gold (#D4AF37), and input fields return to a clean standby state.
3. Make it hold up without the author present	Tested against bad inputs (empty drafts, malformed dates, unsupported file formats), browser refreshes and repeated clicks. Buttons are debounced, toasts never block the screen, and a tampered ledger can be restored in one click.
4. Tighten the interface	Header navigation always stays on one line. A suggested fix is adopted in one click (Accept Suggested Fix), the four-role switcher adapts to context, and dismissing a modal leaves no residual blur.
5. Test the main journeys end to end	An 8-phase live end-to-end test driven by computer use produced 16 timestamped visual receipts (docs/evidence/). It sits alongside 106 passing Pytest tests, a 100% pass rate.
6. Write reproducible documentation	The README guides a judge through cloning, installing and running locally in under 60 seconds, with zero cloud dependencies.

4. The three improvements delivered

Each improvement below was committed at Round 2 and is fully delivered.

4.1 Time-bounded Fact Vault, modality locks and surgical diffing

- **Temporal horizon windows.** Every factual claim carries explicit validity dates [valid_from, valid_until]. A claim that cites an expired fact, such as the 12-week course length in F-DUR-01, is flagged and swapped for the current fact (10 weeks, F-DUR-02).
- **Defence against modality inflation.** Claims are classed as strict, hedged or absolute. Marketing inflation that is not authorised is flagged, for example when “save up to 30%” becomes “guarantees 30% savings”.
- **Surgical character diffing.** Only the offending span is highlighted and replaced ([-12 weeks-] {+10 weeks+}), so the author's voice and all protected words stay untouched.

4.2 Multi-role governance, override-rate telemetry and the SHA-256 ledger

- **Four-role access control.** Creator, Brand Manager, Authorised Reviewer and Brand Admin each see a view suited to their job, with fine-grained permission gating.
- **Reviewer Override Rate (Φ ROR).** Compares reviewer overrides with automated findings to measure policy friction and to point at ambiguous guideline clauses.
- **Cryptographic hash chain.** Submissions, waivers, approvals and fact changes are each signed and linked: $hash_n = SHA\text{-}256(hash_{n-1} \,||\, payload)$.
- **Live tamper defence.** Out-of-band changes to a stored payload are detected immediately, and a clean chain can be restored in one click.

4.3 Multimodal vision maths and accessibility engine

- **CIEDE2000 colour distance.** Perceptual distance ($\Delta E00 \leq 2.5$) is computed in CIE L*a*b* space to catch palette deviations from the brand.
- **WCAG 2.1 AA contrast.** Relative luminance ratios are checked on every text overlay: at least 4.5:1 for body copy and at least 3.0:1 for large display headings.
- **OCR legibility and logo aspect ratio.** Logos must hold a 4.0:1 ratio within ±5%, and detected text bounding boxes are drawn directly on the canvas preview.

5. Worked examples and formulas

5.1 Two claims through the verifier

The two cases below are the ones the product is built to catch. Both come from the Fact Vault behaviour described in section 4.1.

Step	Stale fact	Modality inflation
Draft says	“The course runs 12 weeks.”	“Our plan guarantees 30% savings.”
Vault lookup	Fact F-DUR-01 (12 weeks) has expired; F-DUR-02 (10 weeks) is active.	The approved fact is hedged: savings of up to 30%.
Finding	Stale claim, tied to the expired fact ID.	Modality inflation: a hedged fact used as an absolute claim.
Surgical fix	[-12 weeks-] {+10 weeks+}	Restore the hedge so the wording matches the approved fact.
What is preserved	Everything except the offending span.	The rest of the sentence and the author's voice. Both fixes are appended to the SHA-256 ledger.

5.2 The formulas

- **Intent Preservation Metric.** For a sentence S and a repair S′: IPM = 0.65 × cosine(Embed(S), Embed(S′)) + 0.35 × (1 − WordEditDistance(S, S′) / Length(S)). A suggestion is shown only when IPM is at least 0.85.
- **Reviewer Override Rate.** ROR(rule) = dismissed flags / total flags. Many flags with a low ROR point to a creator training gap; many flags with a high ROR point to an unclear guideline clause.
- **Modality inflation.** A claim is a tuple of subject, predicate, metric and modality. If the matching fact is hedged and the draft is absolute, a deterministic rule raises the flag, and no model decides this step.
- **Hash chain.** $hash_n$ = SHA-256($hash_{n-1}$ || payload).
- **Colour and contrast thresholds.** ΔE00 ≤ 2.5; contrast at least 4.5:1 for body copy and 3.0:1 for large headings; logo aspect ratio 4.0:1 within ±5%.

6. Technical architecture: the nine-layer pipeline

VeriLoom avoids the brittle single-prompt wrapper. It combines deterministic AST compilers with a lightweight, locally hosted BERT-NLI model, in nine layers.

Layer	Responsibility
1. Document ingestion and SHA-256 digest	Reads .txt, .md, .docx and scanned .pdf files and produces a canonical SHA-256 digest.
2. Privacy and sandbox gate	Applies OWASP LLM01 injection defence and masks personal data locally, with no data sent outside.
3. Deontic Brand-AST compiler	Compiles guidelines into abstract syntax trees using modal logic (MUST, MUST_NOT, SHOULD).
4. Time-bounded Fact Vault	Filters institutional knowledge by review date and temporal horizon.
5. Atomic claim verifier	Extracts claim tuples and tests them for contradiction and modality inflation.
6. Multimodal vision auditor	Computes CIEDE2000 colour distance and WCAG 2.1 AA luminance ratios.
7. Verdict and stability gate	Routes each draft into one of four risk tiers: PASS, ADVISORY, VIOLATION or BLOCKED.
8. Intent-locked surgical fix	Generates minimum-edit replacements with an Intent Preservation Metric (IPM) of at least 0.85.
9. Tamper-evident SHA-256 ledger	Appends signed, immutable audit blocks to the persistent hash chain.

7. Implementation and interface highlights

This section records how the finished product behaves in use, beyond the nine-layer logic described above.

7.1 Completeness

All six core workspace modules and the Copilot are fully functional. There are no placeholder screens and no stubbed APIs, and the complete nine-layer neuro-symbolic pipeline runs behind them.

7.2 Interface and usability

- **Navigation.** The header stays on a single line across 1280 px to 1440 px viewports.
- **Role switching.** A contextual four-role switcher changes the view for Creator, Brand Manager, Authorised Reviewer and Brand Admin.
- **Streaming Copilot.** Replies appear word by word with an active cursor.
- **Feedback.** A 10-second metallic gold progress toast (#D4AF37) reports progress without blocking the screen.
- **Visual style.** Translucent frosted-glass chips (rgba(255,255,255,0.08)) and clean default standby inputs.
- **Fixes in one click.** Surgical rewrites are adopted with Accept Suggested Fix, and dismissing a modal leaves no residual blur.

7.3 Reliability under normal use

- **Automated coverage.** 106 tests across 12 modules all pass.
- **Bad input and empty states.** Empty drafts, malformed dates and invalid file formats are handled cleanly.
- **Rapid use.** Duplicate clicks are debounced, and browser refreshes do not lose state.
- **Tampering.** Out-of-band edits to the audit chain are detected, and a clean chain is restored in one click.

7.4 Key implementation parameters

Area	Parameter	Value
Fact Vault	Validity window per fact	[valid_from, valid_until]
Claim verifier	Modality classes	Strict, hedged, absolute
Vision auditor	Colour distance limit	ΔE00 ≤ 2.5 (CIEDE2000, CIE L*a*b*)
Vision auditor	Text contrast (WCAG 2.1 AA)	At least 4.5:1 body; 3.0:1 large headings
Vision auditor	Logo aspect ratio	4.0:1 within ±5%
Verdict gate	Risk tiers	PASS, ADVISORY, VIOLATION, BLOCKED
Surgical fix	Minimum IPM to show a fix	0.85 (measured 0.96)
Ledger	Hash rule	SHA-256 chain over every audit event
Ingestion	Accepted formats	.txt, .md, .docx, scanned .pdf
Privacy	Data handling	PII masked locally, no external transfer

8. Workflow and responsible design

The product follows the human-in-the-loop principles set out in Round 1. The system finds, explains and proposes; people decide.

8.1 Who does what

Role	What they do in VeriLoom	What the system gives them
Creator	Submits a draft and reviews suggested fixes.	Cited findings and one-click, small, intent-preserving fixes before submission.
Brand Manager	Reviews the queue and compares versions.	Surgical diffs and recurring-issue patterns.

Role	What they do in VeriLoom	What the system gives them
Authorised Reviewer	Handles escalated or risky items and may grant a waiver.	Evidence for each finding, and overrides that feed the ROR measure.
Brand Admin	Owns guidelines, approved facts and assets.	Fact Vault with validity dates and override-rate telemetry to spot unclear rules.

8.2 Safeguards

- **Evidence before verdict.** Findings cite a rule or fact ID, and the benchmark recorded no ungrounded high-risk finding.
- **No silent changes.** Suggested fixes are applied only when a person accepts them.
- **Privacy.** Personal data is masked locally and nothing leaves the system for masking, and submitted text is treated as data, never as instructions.
- **Tamper evidence.** Every submission, waiver, approval and fact change is chained by SHA-256, and any out-of-band edit is detected at once.
- **Resilience.** Bad inputs, refreshes and repeated clicks are handled without data loss.

9. Testing and verification

Results come from three sources: the automated Pytest suite, the 54-run golden benchmark on the Northwind set of 18 pieces, and the live end-to-end suite.

9.1 Golden benchmark results

Measure	Result	Target
Violation detection F1	1.000 (100% precision and recall)	At least 0.850
Three-run stability	100.0% (18 of 18 verdicts identical across runs)	At least 90.0%
High-risk hallucination	0.0% (every finding grounded in a verified rule or fact ID)	—
Intent Preservation Metric (IPM)	0.96 (all protected words retained)	At least 0.85
Engine latency	382.6 ms	At least 45% faster than human review

9.2 Automated and live testing

- **Pytest suite.** 106 of 106 tests pass across 12 test files, a 100% pass rate.
- **Live end-to-end suite.** Eight phases were run headlessly with Playwright, producing 16 visual receipts in docs/evidence/.
- **Ledger security.** More than 360 audit blocks were verified, and simulated tamper attacks were caught instantly.

10. Limitations and honest notes

The figures in this submission should be read with the following limits in mind.

- The golden benchmark uses the author's fictional Northwind Academy kit and 18 pieces written by the author. It shows that the failure modes are handled in this setting; it is not a general error rate for real organisations.
- Image checks cover measurable properties (colour, contrast, alt text, aspect ratio). They do not judge taste or meaning.
- Language-model-style judgment can still be wrong, which is why the product uses risk tiers and keeps humans in the approval path.
- Regulatory coverage and multilingual support are roadmap items (section 11), not part of the current build.
- Cells marked [add result] in section 2.2 are not measured in this submission and are left open rather than estimated.

11. Product roadmap and commercial development plan

Phase and timeline	Key deliverables	Commercial and enterprise impact
Phase 1: Ecosystem plugins (Q1–Q2 2027)	A Figma Code Connect plugin for live design-canvas linting, plus CMS webhooks for Webflow, Contentful and Sanity.	Real-time compliance checks inside the design and writing tools creators already use.
Phase 2: Global regimes (Q3–Q4 2027)	Coverage of 12 statutory advertising bodies (FTC, ASA, ARPP, ACCC, BaFin) and multilingual ASTs (DE, FR, JA).	Automated marketing compliance for multinational enterprises.
Phase 3: Hardware enclaves (2028 onward)	Confidential computing on AWS Nitro Enclaves or Intel SGX, and zero-knowledge succinct proofs (zk-SNARKs).	Verification with zero data exposure for defence, banking and confidential IP work.

12. Reproducing the results

Everything in this document can be checked by a judge.

- **Live product:** https://veriloom-3d0g.onrender.com/
- **Source code:** github.com/nandhak6197-a11y/VeriLoom (main branch).
- **Local run:** the README explains how to clone, install and run in under 60 seconds with no cloud dependency.
- **Test evidence:** 106 Pytest tests across 12 files, and 16 timestamped receipts from the end-to-end run in docs/evidence/.
- **Research and sources:** the Round 1 submission holds the evidence ledger and references.

Links and signature

Live deployment: https://veriloom-3d0g.onrender.com/ · Source: github.com/nandhak6197-a11y/VeriLoom

Team: nandhaashwin111222 · Team leader: Nandha K · Email: nandhaashwin111222@gmail.com

Cryptographic signature: SHA256:nandhaashwin111222:NandhaK:nandhaashwin111222@gmail.com:2026